Inverse Estimation of Multiple Muscle Activations From Joint Moment With Muscle Synergy Extraction

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***Abstract*—Human movement is produced resulting from synergetic combinations of multiple muscle contractions. The resultant joint movement can be estimated through the related multiple-muscle activities, which is formulated as the forward problem. Neuroprosthetic applications may benefit from cocontraction of agonist and antagonist muscle pairs to achieve more stable and robust joint movements. It is necessary to estimate the activation of each individual muscle from desired joint torque(s), which is the inverse problem. A synergy-based solution is presented for the inverse estimation of multiple muscle activations from joint movement, focusing on one degree-of-freedom tasks. The approach comprises muscle synergy extraction via the nonnegative matrix factorization algorithm. Cross validation is performed to evaluate the method for prediction accuracy based on experimental data from ten able-bodied subjects. The results demonstrate that the approach succeeds to inversely estimate the multiple muscle activities from the given joint torque sequence. In addition, the other one's averaged synergy ratio was applied for muscle activation estimation with leave-one-out cross-validation manner, which resulted in 9.3% estimation error over all the subjects. The obtained results support the common muscle synergy-based neuroprosthetics control concept.**

***Index Terms*—Electromyography (EMG), inverse problem, muscle activations, muscle synergy.**

I. Introduction

PATIENTS in need of rehabilitation, assistance, and neuroprosthetics are increasing in our aging society all over the world. Reducing the expected cost for the health care of aging society is really a critical problem [1]. The social benefit that could be derived from advanced computational methods in motor rehabilitation would be quite important, because restoration of muscle motor capability by means of neuroprosthetics could be beneficial to improvement of quality of life for motor-impaired patients during their daily movements. Redundant muscle contractions produce articulated human body movements. Studying motor coordination is essential to investigate its optimal relationship among multiple-muscle activations. In muscle coordination studies, the electrical activities of contracted muscles can be measured by electromyography (EMG), and processed EMG signals are often used to represent muscle activations [2]. Since EMG is highly correlated with the movement of joints, the joint movement induced by muscle groups can be analyzed bilaterally: from forward and inverse perspectives. The forward problem consists of determining the resultant joint torque from the multiple-muscle activations, and in past decades, several contributions were proposed to establish the forward relationship between EMG and joint torque [3]–[7]. For example, Clancy *et al.* [4], [5] summarized several basic forward linear and nonlinear models on multiple extension and flexion EMG signals to estimate elbow joint torque under constant postures. Sinkjaer *et al.* [6] measured both extension and flexion muscles to track the ankle joint torque at different contraction levels. Kent [7] applied the neurogenetic method to establish the nonlinear mapping from EMG to ankle joint torque. Identifying the forward relationship between EMG and joint movement can contribute to certain rehabilitation applications, such as the control of a human-interactive robot [8], prosthetic limbs [9], or exoskeleton [10], by means of an EMG-driven approach to determine the subject's mechanical intention regarding the target motion.

In the domain of neuroprosthetics [11], [12] based on functional electrical stimulation (FES), conversely, the muscle group activities for the joint movement have to be determined to meet the desired joint torque, which is formulated as the inverse problem. The inverse problem involves indefinite solutions because of the inherent muscle redundancy with respect to the associated joints, and there is no bijection between one degree of freedom (DOF) and the cocontracted muscles spanning around the joint. The database technique could be applied as a trial to recover/estimate muscle activity from the motion [13]. For such an objective, muscle synergy is a useful and insightful concept to investigate cocontractions and the coordination pattern among them [14]. In a physiological sense, muscle synergy reveals how the central nervous system (CNS) elicits redundant muscle groups in a coordinated manner. It can be obtained from multiple-muscle activations via dimension reduction techniques, aiming at diminishing a higher dimensioned signal space to a lower dimensioned feature space.

In general, most current FES systems activate one agonist muscle group for each DOF. However, instead of stimulating only one muscle, it would help to minimize the rapid fatigue that is a common clinical problem in FES systems if the necessary force to be managed could be distributed by activating multiple muscles. To assist the future design of FES systems with distributed control over multiple muscles, information on muscle synergy could be exploited to naturally follow the way CNS regulates and optimally take advantage of muscle redundancy [15]–[18]. By extracting muscle synergy information from healthy subjects and simultaneously measuring the corresponding

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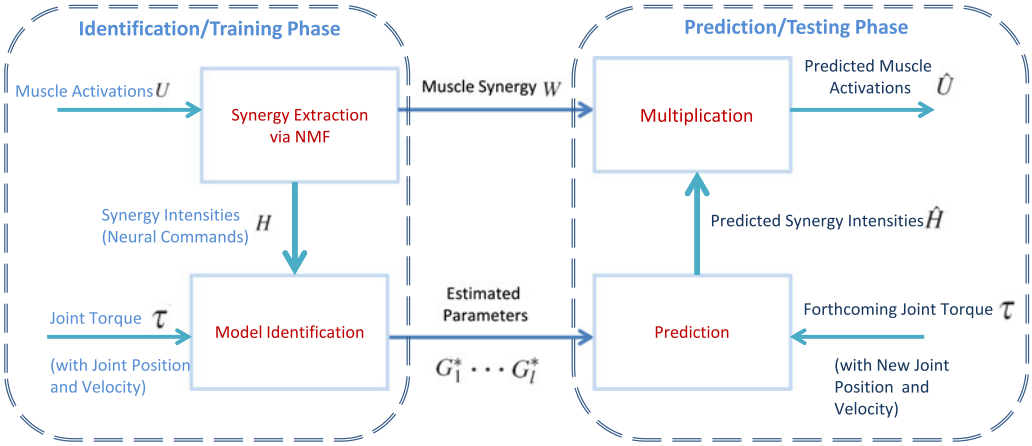


Fig. 1. Block schematic diagram of the process for inverse estimation of multiple-muscle activations from the joint movement.

kinetics/kinematics, insight into individual muscle's contribution to the joint flexors/extensors would be obtained. The synergies extracted from healthy subjects, together with their joint moment data, could then be used to design stimulation patterns for multiple muscles of incomplete spinal cord injured subjects who still have remaining volitional contractions of their partially deficient muscles [18]. The recruitment order in percutaneous FES is in a nonphysiological manner, which induces muscle fatigue faster than natural voluntary contractions [19]. Employing multiple redundant muscle activations in FES can promote reducing fatigue of the stimulated muscles and increasing joint articulation stability by using the cocontraction of antagonist muscles [20]. An insightful view of how multiple-muscle activations are coordinated toward the desired joint torque sequence in able-bodied subjects would help to reduce the time-consuming empirical FES tuning on paralyzed patients [21]. Thus, the computational inverse estimation of multiple-muscle activations is of broad interest for FES applications and should help to further elucidate the underlying physiological processes, which is the primary motivation for this study.

In our preliminary works [22], [23], an efficient multiple-regression-based method was proposed for the inverse estimation of the three muscle activations from ankle joint movement. Such an approach estimates multiple-muscle activations separately without extracting synergy patterns. However, to further investigate the natural muscle cocontraction patterns, it may be necessary to extract the muscle synergy of agonist and antagonist muscle groups under different joint flexions/extentions during both isometric and isotonic contractions. "Isometric" means that muscle contracts when the joint angle remains unchanged, and "isotonic" refers to a muscle contraction that is associated with the change in joint angle (muscle length) with equal tension. The motion under lifting an object is an example of isotonic contractions, which are often employed in our daily life. In this paper, we propose a synergy-based framework for inverse estimation of multiple-muscle activations from joint torque. The method comprises two consequent procedures: muscle synergy extraction and polynomial model identification between synergy intensities and joint torque. One could control the joint torque at different muscle activation levels if there is prior knowledge of the muscle synergies, which suggests a synergy-assisted estimator for joint torque control by FES as a potential application. Experimental validation is achieved with ten able-bodied subjects who performed random voluntary contractions around the ankle joint in isometric and isotonic conditions. This method successfully provides inverse estimations of the three muscle activations from the ankle joint, in addition to muscle cocontraction information during plantar and dorsi flexions.

The organization of the work is as follows. The experimental setup and the approach for inverse estimation of muscle activations are addressed in Section II. The validation results and discussion are presented in Section III. Detailed discussions are presented in Section IV. Section V draws the final conclusions.

II. Materials and Methods

The general schematic diagram of the entire inverse estimation process is illustrated in Fig. 1. First, the muscle synergy and the synergy intensities are extracted through nonnegative matrix factorization (NMF) from multiple-muscle activations. Second, identification of the model is performed between the extracted synergy intensities and the corresponding joint movement. For prediction purposes (cross validation), only the joint torque information (additionally with joint position and velocity information in nonisometric case) is used to predict the muscle activations with the identified model.

A. Experimental Setup and Data Processing

The EMG and the joint torque data were acquired from ten able-bodied subjects (seven males and three females, 28.5 ± 3 years old, 67.5 ± 8.8 kg weight, and 170.7 ± 6.1 cm height) upon their consent. The study was approved by the ethical committee for persons' protection managed by CHRU Montpelier (2011-A01033-38), dedicated to the study of the lower limb biomechanics within ANR SoHuSim project. The experiment setup is shown in Fig. 2. The subjects were seated on a chair with their right foot attached to a Biodex Dynamometer (Biodex Medical Systems Inc., Shirley, NY, USA). The setting was 90° for the

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Fig. 2. Experimental setup for one of the subjects. The electrodes of the EMG acquisition system are placed on the muscles (SOL, MG, and TA). Each EMG is acquired synchronously with ankle torque measurement.

ankle joint and 110° for the knee joint, while the joint center was aligned with the axis of the dynamometer. Straps were used on the pelvis and shoulders to secure subjects' position on the chair. Thigh movement was also restricted by a strap. The EMG electrodes were placed over the muscle belly of the SOL (Soleus), MG (Medial Gastrocnemius), and TA (Tibialis) muscles according to the SENIAM recommendation [24] of electrodes placement based on crus anatomy. The bipolar AgCl EMG electrodes (round with 2-cm radius) were aligned along the direction of muscle fiber with 20-mm interelectrode spacing. The reference electrode was placed on the patella of the contralateral leg. The subjects were told to perform maximum voluntary contractions (MVC) in both plantar- and dorsi-flexion directions for two times. They were then asked to voluntarily make the ankle plantar-flexion and dorsi-flexion movement randomly including both directional contractions and both fast-short (1–2 s) and slow-long (>2 s) contractions. The random session which last approximately for 2–3 min and contained about 8–20 random contractions, was repeated two times in isometric and another two times in isotonic conditions. Synchronous acquisition of the joint torque and differential EMG signals were performed with a sample frequency of 2048 Hz by the EMG100c amplifier (gain 1000) and the Biopac MP100 system (Biopac Systems, Inc., Santa Barbara, CA, USA). The Biopac EMG100c hardware has a high-pass filter with a 1-Hz cutoff frequency. After acquisition, the raw EMG signals were rectified and low-pass filtered with 2-Hz cutoff frequency. Next, the processed EMG signals which were scaled against the muscles' own MVC values, respectively, were normalized to represent the muscle activations $U \in [01]$. The data acquired during each experiment session were divided into two phases with ratio approximately being 2:3, i.e., one is the identification (training) phase and the other is the prediction (testing) phase. Each phase contains a series of fast-short and slow-long contractions randomly generated by the subject(s).

B. Muscle Synergy Extraction

Consider the multiple-muscle activations u_{SOL}, u_{MG}, and u_{TA} of the SOL, MG, and TA muscles during the identification time period $[0, t_{id}]$ (Here, notations with these muscles are used for demonstration, but they can be any other muscle group combination involved in a joint.)

$$U = \begin{bmatrix} u_{\mathrm{SOL}} \\ u_{\mathrm{MG}} \\ u_{\mathrm{TA}} \end{bmatrix} = \begin{bmatrix} u_{\mathrm{SOL}}(0) & u_{\mathrm{SOL}}(1) & \dots & u_{\mathrm{SOL}}(t_{id}) \\ u_{\mathrm{MG}}(0) & u_{\mathrm{MG}}(1) & \dots & u_{\mathrm{MG}}(t_{id}) \\ u_{\mathrm{TA}}(0) & u_{\mathrm{TA}}(1) & \dots & u_{\mathrm{TA}}(t_{id}) \end{bmatrix}. \tag{1}$$

One can use NMF to decompose U as [14]

$$U = WH + E \approx WH \tag{2}$$

where W denotes the muscle synergy ratio matrix, H denotes the extracted synergy intensity vector (neural command), and E is the decomposition residual error term. The decomposition for updating entries h_{lm} and w_{nl} of H and W is conducted with the following iterative algorithm [14]:

$$h_{lm} \leftarrow h_{lm} \frac{[W^T U]_{lm}}{[W^T W H]_{lm}} \tag{3}$$

$$w_{nl} \leftarrow w_{nl} \frac{[U H^T]_{nl}}{[W H H^T]_{nl}}. \tag{4}$$

It should be noted here that there are two types of articulation for ankle joint sagittal movement for one DOF: plantar flexion and dorsi flexion [6], [20]. Therefore, the number of muscle synergies can be finalized as two for one DOF corresponding to the flexion and extension of the joint, that is, the dimension of the reduced resultant subspace is chosen as $l = 2$ in this study.

C. Identification

Different joint torque levels can be achieved by actuating different levels of muscle synergy intensities. To do so, identification is first established between the joint torque and the synergy intensities. In the literature [4], [5], [25], activation-to-torque (u_i to τ) forward mapping is described in a general nonlinear form $f_i(\cdot), i \in \{\mathrm{SOL}, \mathrm{MG}, \mathrm{TA}\}$ as

$$\tau = \sum_i f_i(u_i).$$

It implies that a nonlinear relationship from torque to activations exists. Throughout this paper, we assume that this inverse relationship can be expressed as the following general nonlinear form $g_i(\cdot)$:

$$u_i = g_i(\tau).$$

According to (2) and the nonlinear relationship above, the activations of the three muscles can be rewritten as

$$\begin{bmatrix} u_{\mathrm{SOL}} \\ u_{\mathrm{MG}} \\ u_{\mathrm{TA}} \end{bmatrix} = \begin{bmatrix} g_{\mathrm{SOL}}(\tau) \\ g_{\mathrm{MG}}(\tau) \\ g_{\mathrm{TA}}(\tau) \end{bmatrix} \approx W \begin{bmatrix} H_1 \\ \vdots \\ H_l \end{bmatrix}. \tag{5}$$

Analytically, the extracted synergy intensity vector is

$$\begin{bmatrix} H_1 \\ \vdots \\ H_l \end{bmatrix} \approx W^{\dagger} \begin{bmatrix} g_{\mathrm{SOL}}(\tau) \\ g_{\mathrm{MG}}(\tau) \\ g_{\mathrm{TA}}(\tau) \end{bmatrix} \tag{6}$$

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where $W^{\dagger}$ is the Moore–Penrose pseudoinverse [26] of muscle synergy matrix W. Consequently, each individual synergy intensity vector can be estimated generally as follows:

$$\begin{aligned} \tilde{H}_1 &= G_1^* \Phi_{id} \\ &\vdots \\ \tilde{H}_l &= G_l^* \Phi_{id} \end{aligned} \tag{7}$$

where

$$\Phi_{id} = \begin{bmatrix} \tau_0 & \tau_0 & \tau_0 & \dots & \tau_0 \\ \tau(0) & \tau(1) & \tau(2) & \dots & \tau(t_{id}) \\ \tau^2(0) & \tau^2(1) & \tau^2(2) & \dots & \tau^2(t_{id}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \tau^q(0) & \tau^q(1) & \tau^q(2) & \dots & \tau^q(t_{id}) \end{bmatrix}$$

in isometric case

$$or\ \ \Phi_{id} = \begin{bmatrix} \tau_0 & \tau_0 & \tau_0 & \dots & \tau_0 \\ \tau(0) & \tau(1) & \tau(2) & \dots & \tau(t_{id}) \\ \tau^2(0) & \tau^2(1) & \tau^2(2) & \cdots & \tau^2(t_{id}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \tau^q(0) & \tau^q(1) & \tau^q(2) & \dots & \tau^q(t_{id}) \\ p(0) & p(1) & p(2) & \cdots & p(t_{id}) \\ v(0) & v(1) & v(2) & \dots & v(t_{id}) \end{bmatrix}$$

in isotonic case

and the estimated parameters are

$$\begin{aligned} G_1^* &= H_1 \Phi_{id}^{\dagger} \\ &\vdots \\ G_l^* &= H_l \Phi_{id}^{\dagger} \end{aligned} \tag{8}$$

where $\Phi_{id}^{\dagger}$ is the Moore–Penrose pseudoinverse of Φ_{id}, τ_0 denotes the baseline joint torque value, and $p(\cdot)$ and $v(\cdot)$ denote the joint position and velocity information, respectively.

D. Prediction of Muscle Activations

The parameters $G_1^*, \dots, G_l^*$ of the nonlinear model of the relationship between muscle synergy intensities and joint torque are determined in the identification phase. During the prediction phase within the prediction time period $[t_{id}+1, t]$, only the forthcoming joint torque information (with joint position and velocity information) is used to assess the model's ability to predict, which serves to cross validate the identified model. The predicted individual synergy intensities are obtained by

$$\begin{aligned} \hat{H}_1 &= G_1^* \Phi_{pr} \\ &\vdots \\ \hat{H}_l &= G_l^* \Phi_{pr} \end{aligned} \tag{9}$$

where

$$\Phi_{pr} = \begin{bmatrix} \tau_0 & \tau_0 & \dots & \tau_0 \\ \tau(t_{id}+1) & \tau(t_{id}+2) & \dots & \tau(t) \\ \tau^2(t_{id}+1) & \tau^2(t_{id}+2) & \dots & \tau^2(t) \\ \vdots & \vdots & \ddots & \vdots \\ \tau^q(t_{id}+1) & \tau^q(t_{id}+2) & \dots & \tau^q(t) \end{bmatrix}$$

in isometric case

$$or\ \ \Phi_{pr} = \begin{bmatrix} \tau_0 & \tau_0 & \dots & \tau_0 \\ \tau(t_{id}+1) & \tau(t_{id}+2) & \dots & \tau(t) \\ \tau^2(t_{id}+1) & \tau^2(t_{id}+2) & \dots & \tau^2(t) \\ \vdots & \vdots & \ddots & \vdots \\ \tau^q(t_{id}+1) & \tau^q(t_{id}+2) & \dots & \tau^q(t) \\ p(t_{id}+1) & p(t_{id}+2) & \dots & p(t) \\ v(t_{id}+1) & v(t_{id}+2) & \dots & v(t) \end{bmatrix}$$

in isotonic case.

According to the original muscle-activation decomposition formulation (2), the predicted muscle activations for each individual subject can be written as

$$\hat{U} \approx W \hat{H}. \tag{10}$$

The ability to deal with the redundant muscle combinations would have practical application in the neuroprosthetic system control. It is important to know how the multiple-muscle activations should be inversely generated quantitatively to follow the necessary joint torque sequence (with joint position and velocity information) depending on the target task, using the individually identified model.

III. Results

In this section, we present the inverse estimation/prediction results obtained with the aforementioned approach, based on the experimental data collected from the ten able-bodied subjects. In the identification/training phase, the muscle synergies and intensities are extracted during the identification time period $[0, t_{id}]$ through the NMF algorithm. The parameters G_1^*, G_2^*, which map ankle joint torque and the synergy intensities, are then estimated by equation (8). In the prediction/testing phase, the muscle activations are further predicted with the forthcoming ankle joint torque information as the sole input. The evaluation is made under both isometric and isotonic conditions, in order to present the prediction performances in both static and dynamic situations.

The muscle synergies for all ten subjects are extracted from the muscle activation recordings in the corresponding pair of plantar and dorsi flexions. To quantitatively evaluate each muscle's contribution to the resultant torque of these plantar and dorsi flexions, the normalized synergy ratio for each synergy pair l is defined as follows.

$$\begin{aligned} \text{Ratio}_{\text{SOL}} &= \frac{w_{1l}}{w_{1l} + w_{2l} + w_{3l}} \\ \text{Ratio}_{\text{MG}} &= \frac{w_{2l}}{w_{1l} + w_{2l} + w_{3l}} \\ \text{Ratio}_{\text{TA}} &= \frac{w_{3l}}{w_{1l} + w_{2l} + w_{3l}}. \end{aligned}$$

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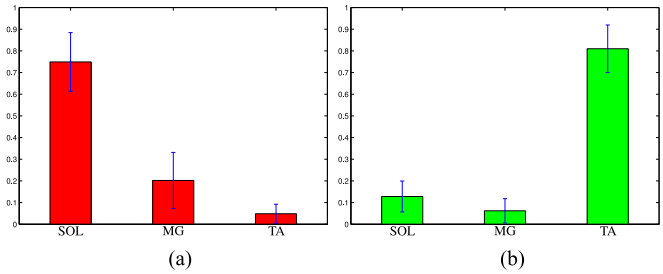


Fig. 3. Obtained muscle synergies with standard deviation for the ten able-bodied subjects. It represents the means of normalized muscle synergy ratios. (a) Synergy ratio pattern at plantar flexion. (b) Synergy ratio pattern at dorsi flexion.

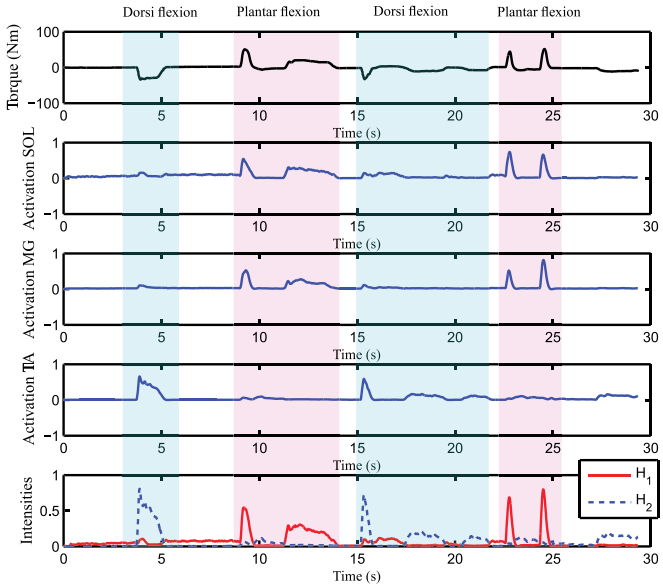


Fig. 4. From upper to lower: (a) ankle joint torque, (b) SOL activation, (c) MG activation, (d) TA activation, and (e) extracted synergy intensity components H_1 and H_2 from muscle activations U by NMF. Note the random pattern of muscle activations between the positive torque period and the negative period, which correspond, respectively, to plantar flexion and dorsi flexion.

Fig. 3 shows the mean values with standard deviations of the normalized synergy ratios of the ten subjects, with two distinct types of muscle synergy of the three muscles for the subjects.

Fig. 3(a) shows that SOL is the main agonist muscle conducting plantar flexion of the ankle joint, whereas the Fig. 3(b) shows that TA is the main agonist muscle for dorsi flexion of the ankle joint. The cocontractions of agonist and antagonist muscles can be observed for both directions of flexion. For more detailed observation, we take Subject V1 as an illustrative example as shown in Fig. 4. The two synergy intensity (neural command) vectors H_1 and H_2 extracted from the muscle activations of the three muscles correspond to Synergy 1 and 2, respectively. In Fig. 4, one can intuitively observe that, the positive ankle joint torque (i.e., plantar-flexion torque) highly correlates with synergy intensity H_1, and the negative ankle joint torque (i.e., dorsi-flexion torque) highly correlates with synergy intensity H_2. Since the extracted synergy intensity is the direct reflection of neural command [14], it may indicate that the CNS may dispatch two different neural commands to control the joint extension and flexion by, respectively, activating two different pairs of muscle combinations.

To assess the prediction/testing performance of this synergy-based approach, the inverse estimation for multiple-muscle activation is demonstrated by cross validation in the ten subjects. In our previous works [22], [23], the multiple regression method was efficient for the inverse estimation of multiple-muscle activations. Multiple-regression analysis is without dimension reduction. To compare evaluation between the synergy-based method and the aforementioned multiple regression method, the prediction results using the multiple regression method [22], [23] are presented here as well. To assess the global performance of prediction (cross validation), we define the following root-mean-square errors (RMSEs) and variance (var) accounted for (VAF) as follows as the performance indexes:

$$\text{RMSE}_{id} = \sqrt{\frac{\|\tilde{U}_i - U_{i,\text{m}}\|^2}{N_{id}}}, \text{RMSE}_{pr} = \sqrt{\frac{\|\hat{U}_i - U_{i,\text{m}}\|^2}{N_{pr}}}$$

$$\text{VAF} = 1 - \frac{var(\hat{U}_i - U_{i,\text{m}})}{var(U_{i,\text{m}})}, \quad i \in \{\text{SOL},\ \text{MG},\ \text{TA}\}$$

where $U_{i,\text{m}}$ denotes the measured activation vectors of muscle i, and N_{id} and N_{pr} are denoted as the sampling numbers during the identification and prediction phases, respectively.

Fig. 5 illustrates the isometric case for Subject V1. The identification is finished in a time period ranging 0 s to around 47 s, and from 47 s onward, only the identified model and the forthcoming ankle joint torque information are provided for muscle activation prediction. In Fig. 5, we see that the predicted muscle activations of the three muscles fit well with the measured ones from time instant 47 s to the end. Even when dorsi flexion and plantar flexion are contained randomly with short (1–2 s) and long (>2 s) term contractions, the proposed method is able to produce the resultant multiple-muscle activations from the ankle joint movement with good accuracy. Table I presents a more detailed investigation, with the RMSEs and VAFs for the muscle activation estimation via the synergy-based method, and the multiple regression method [22], [23] under the isometric condition. For cross validation of the synergy-based method under the isometric condition, the RMSE is about 4.03 ± 2.31 and the VAF is about 82.60 ± 13.25, while for that of the multiple regression method the RMSE is about 4.20 ± 2.60 and the VAF is about 82.15 ± 15.00. From this comparison, we observe that the synergy-based method inversely estimates the multiple-muscle activations successfully as the multiple regression method.

The p-value obtained by similarity analysis through cross-correlation measure for the errors of the two approaches is less than 0.05, indicating that they are with high similarity. This indicates that the synergy-based method possesses the same estimation quality as the multiple regression method regardless of dimension reduction after muscle synergy extraction. We also use these two methods to estimate the three muscle activations under the isotonic condition. In this situation, the joint position and velocity are also involved for inverse estimation to take into

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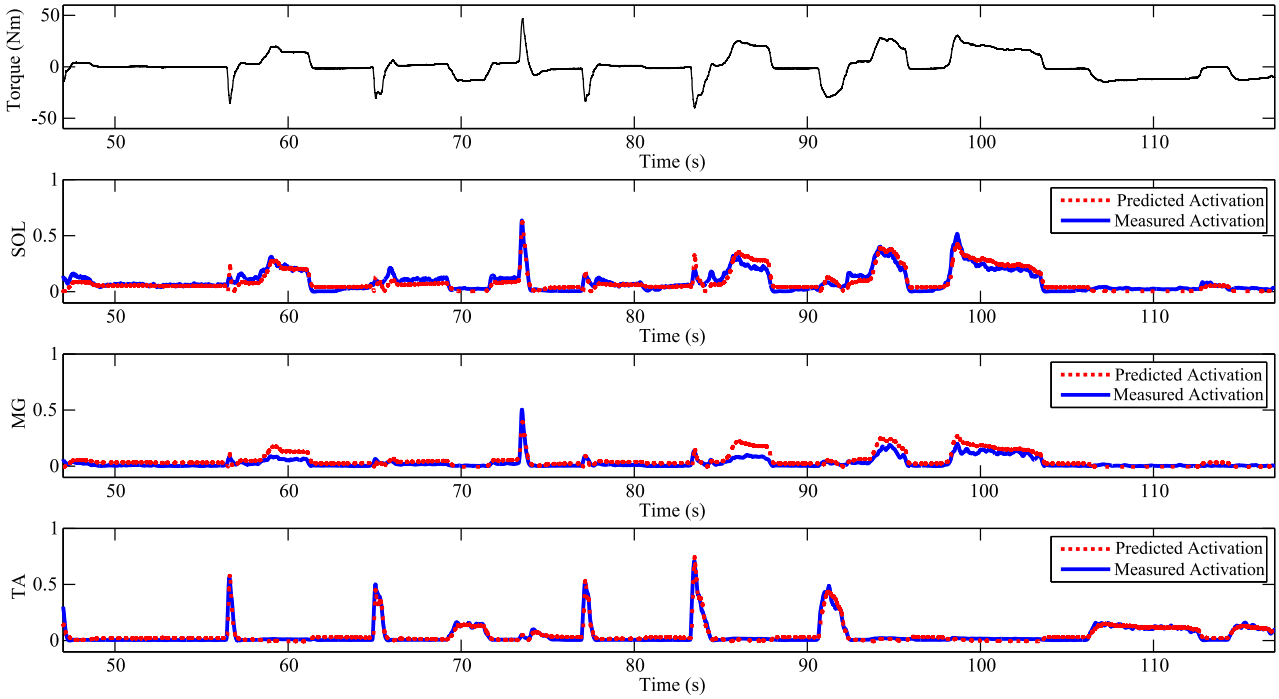


Fig. 5. Cross validation for Subject V1: From around $t = 47$ s, the prediction is conducted with the identified model and the incoming joint torque. Red-dotted curves represent the predicted activations and blue solid curves represent the measured activations. Torque below and above the baseline represents the dorsi flexion and plantar flexion, respectively.

TABLE I
RMSE (%) AND VAF (%) OF ESTIMATED MUSCLE ACTIVATIONS OF THE TEN SUBJECTS IN ISOMETRIC CONDITION

Method		Synergy-based method			Multiple regression method [22], [23]		
Profile		$RMSE_{id}$	$RMSE_{pr}$	VAF	$RMSE_{id}$	$RMSE_{pr}$	VAF
	SOL	4.14	3.82	82.78	3.75	3.83	82.75
V1	MG	3.83	3.83	58.36	2.67	3.08	58.66
	TA	3.18	2.68	92.70	3.18	2.65	92.80
	SOL	6.55	5.47	70.13	6.55	5.69	65.51
V2	MG	2.51	1.90	66.80	2.49	1.55	75.27
	TA	4.55	4.19	93.45	4.55	4.15	93.55
	SOL	3.31	4.27	79.66	2.95	4.03	83.80
V3	MG	7.32	4.54	72.26	6.82	4.97	34.68
	TA	3.33	3.03	91.84	3.29	2.90	92.62
	SOL	4.95	5.09	83.94	5.11	5.41	81.33
V4	MG	3.73	3.77	84.98	3.82	4.10	81.45
	TA	3.82	6.07	94.93	3.27	6.53	94.54
	SOL	3.14	3.16	67.09	3.14	3.16	61.09
V5	MG	0.49	0.50	71.13	0.41	0.45	75.55
	TA	4.69	7.49	95.98	4.69	7.49	95.99
	SOL	7.39	7.73	60.72	6.85	9.29	57.21
V6	MG	5.36	5.36	45.37	3.33	5.32	63.98
	TA	5.19	5.19	93.72	5.10	5.87	93.66
	SOL	4.55	4.79	90.56	4.55	4.80	90.56
V7	MG	0.87	0.77	84.32	0.86	0.69	85.92
	TA	0.85	0.90	94.80	0.85	0.90	94.79
	SOL	0.58	0.63	90.74	0.58	0.63	90.76
V8	MG	0.16	0.31	73.30	0.16	0.29	76.14
	TA	1.96	2.04	94.29	1.96	2.04	94.28
	SOL	5.86	6.56	86.72	4.72	8.22	82.26
V9	MG	6.02	3.16	93.43	5.93	3.66	94.12
	TA	3.07	4.02	95.80	3.06	3.82	96.42
	SOL	5.03	9.09	88.31	4.88	8.04	90.89
V10	MG	3.23	7.71	86.16	2.41	9.61	89.61
	TA	2.66	3.03	93.86	2.66	2.92	94.32
Overall Average		3.74 ± 1.96	4.03 ± 2.31	82.60 ± 13.25	3.48 ± 1.85	4.20 ± 2.60	82.15 ± 15.00

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TABLE II
RMSE (%) AND VAF (%) OF ESTIMATED MUSCLE ACTIVATIONS OF THE TEN SUBJECTS IN ISOTONIC CONDITION WITH CONSTANT PLANTAR LOAD 10 N · M AND DORSI LOAD 5 N·M*

Method		Synergy-based method			Multiple regression method [22], [23]		
Profile		$RMSE_{id}$	$RMSE_{pr}$	VAF	$RMSE_{id}$	$RMSE_{pr}$	VAF
	SOL	1.44 (1.53)	4.90 (5.12)	79.44 (77.75)	1.43 (1.53)	4.93 (5.12)	79.15 (77.77)
V1	MG	0.98 (1.15)	4.59 (4.83)	44.14 (37.99)	0.95 (1.15)	4.52 (4.84)	45.74 (37.72)
	TA	2.28 (2.96)	5.66 (6.90)	92.66 (89.13)	2.47 (3.21)	6.14 (7.48)	92.66 (89.14)
	SOL	7.86 (7.90)	16.69 (16.69)	61.69 (61.96)	7.83 (7.89)	16.66 (16.65)	61.66 (62.06)
V2	MG	5.64 (5.72)	11.83 (11.91)	56.07 (55.53)	5.52 (5.70)	11.61 (11.91)	57.68 (55.48)
	TA	7.86 (10.84)	12.28 (14.97)	78.03 (68.24)	7.80 (10.83)	12.29 (14.98)	77.99 (68.19)
	SOL	3.17 (3.67)	4.22 (3.66)	59.65 (71.97)	2.99 (3.52)	3.78 (3.17)	65.58 (76.65)
V3	MG	4.77 (5.54)	6.55 (6.74)	75.14 (74.34)	4.65 (5.44)	6.58 (6.74)	77.62 (77.21)
	TA	8.72 (10.73)	11.63 (12.71)	81.86 (78.45)	8.72 (10.73)	11.53 (12.28)	82.00 (78.59)
	SOL	5.18 (5.29)	7.33 (7.57)	70.54 (69.14)	5.14 (5.29)	7.42 (7.50)	69.66 (69.50)
V4	MG	4.01 (3.97)	5.97 (5.91)	69.09 (69.82)	3.93 (3.96)	5.74 (5.98)	71.57 (69.27)
	TA	4.40 (5.49)	5.79 (5.04)	87.35 (90.05)	4.40 (5.49)	5.79 (5.03)	87.31 (90.06)
	SOL	14.03 (14.19)	12.95 (13.12)	87.33 (86.88)	14.76 (14.96)	13.17 (13.44)	85.37 (84.84)
V5	MG	3.25 (3.10)	3.59 (3.47)	84.45 (85.43)	2.56 (2.71)	3.46 (3.57)	85.19 (83.83)
	TA	9.57 (11.12)	12.17 (12.50)	92.68 (91.77)	12.26 (13.44)	20.10 (21.85)	92.57 (91.50)
	SOL	9.39 (11.27)	10.34 (10.63)	64.30 (63.97)	9.36 (11.83)	10.71 (10.94)	56.07 (55.11)
V6	MG	9.01 (9.63)	7.14 (7.58)	81.77 (79.18)	9.28 (10.21)	7.52 (7.61)	81.02 (80.98)
	TA	6.52 (16.66)	6.35 (11.67)	80.68 (35.39)	6.68 (17.10)	6.23 (12.40)	82.72 (27.91)
	SOL	2.94 (3.03)	4.63 (4.77)	52.64 (51.92)	2.94 (3.03)	4.65 (4.80)	53.51 (52.16)
V7	MG	3.09 (3.13)	4.31 (4.36)	43.43 (43.09)	3.10 (3.13)	4.27 (4.31)	42.90 (42.83)
	TA	3.00 (3.65)	3.16 (3.88)	78.49 (63.50)	3.02 (3.67)	3.29 (4.00)	76.47 (61.12)
	SOL	0.55 (0.62)	0.86 (0.92)	80.69 (78.62)	0.56 (0.63)	0.89 (0.93)	79.53 (77.90)
V8	MG	0.34 (0.35)	0.20 (0.21)	79.87 (77.78)	0.33 (0.35)	0.20 (0.21)	81.06 (77.56)
	TA	1.77 (1.86)	2.22 (2.25)	96.11 (95.80)	1.79 (1.87)	2.18 (2.23)	96.21 (95.87)
	SOL	8.27 (9.81)	16.07 (17.65)	49.43 (36.79)	8.15 (9.78)	16.43 (18.06)	47.27 (34.16)
V9	MG	10.04 (12.48)	14.28 (16.39)	68.02 (55.25)	9.94 (12.46)	13.87 (15.98)	69.73 (57.21)
	TA	2.49 (2.57)	3.61 (3.61)	90.92 (90.92)	2.18 (2.57)	3.55 (3.56)	91.19 (91.17)
	SOL	4.54 (4.55)	5.45 (4.86)	74.82 (79.66)	5.47 (5.48)	5.50 (5.46)	73.87 (74.25)
V10	MG	6.14 (7.28)	8.00 (8.46)	89.88 (88.34)	6.75 (7.56)	8.17 (9.12)	89.43 (86.63)
	TA	3.26 (3.46)	3.85 (4.13)	93.49 (92.53)	3.46 (3.72)	4.22 (4.45)	92.35 (91.46)
Overall Average		5.15 ± 3.34 (6.11 ± 4.34)	7.23 ± 4.42 (7.73 ± 4.88)	74.82 ± 15.07 (71.37 ± 17.68)	5.28 ± 3.59 (6.30 ± 4.57)	7.51 ± 4.94 (8.15 ± 5.48)	74.83 ± 15.03 (70.60 ± 18.44)

*Figures inside parentheses are without joint position and velocity information and ones outside parentheses are with joint position and velocity information.

account the muscle response changes due to the force-length-velocity relationship. Table II shows that the performances of two methods in the isotonic case are similar, indicating that the synergy-based method is similar in estimation quality to the multiple regression method under the isotonic condition (p-value < 0.05 by similarity analysis). In Table II, the values within parentheses indicate the inverse estimation performance results without joint position and velocity information, whereas the values outside the parentheses are with the joint position and velocity information taken into account. We see that inverse estimation is improved for both methods by including ankle joint position and velocity information in the isotonic situation (i.e., smaller RMSEs and larger VAFs). Although additional joint position and velocity information could improve the inverse estimation, no significant statistic difference between means is observed (p-value > 0.05 by one-way analysis of variance (ANOVA)).

Table III shows the leave-one-out cross validation, by which the average synergy of the other nine subjects is used for cross validation in the isotonic case to evaluate the intersubject differences on synergy pattern. Each of the subjects adopted the other subjects' normalized average synergy ratios in their own cross validation, respectively, and from Table III, we observe that, although the performance tends to be degraded in this leave-one-out cross validation, the difference in accuracy is not so large when it is compared to Table II.

IV. DISCUSSION

The proposed framework involves the muscle synergy extraction of multiple-muscle activations for inverse estimation from joint torque. Within this framework, NMF is applied to extract muscle synergies from multiple-muscle activations to obtain the muscle cocontraction patterns corresponding to different levels and directions of joint flexion/extension. In this study, two synergy patterns are selected to account for the two types of articulation (plantar and dorsi flexions) for one-DOF sagittal movement. Similarly, for the flexion and extension of other joints that mainly focus on one-DOF tasks like the knee or elbow, two synergy patterns can be used to divide the muscle groups into flexor and extensor muscles in a systematic manner. The extracted synergy intensity vectors may help to analyze the individual correlations and the driving impact to the joint space. Fig. 4 shows that each component of synergy intensity

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TABLE III
RMSE (%) AND VAF (%) OF INVERSE ESTIMATION BY THE SYNERGY-BASED METHOD IN ISOTONIC CONDITION: EVERY OTHER NINE SUBJECTS' NORMALIZED AVERAGE SYNERGY RATIOS ARE APPLIED TO REPLACE THE REST ONE'S OWN SYNERGY RESPECTIVELY FOR LEAVE ONE-OUT CROSS VALIDATION

Method		Synergy-based method	
Profile		$RMSE_{pr}$	VAF
	SOL	7.89	50.75
V1	MG	3.72	73.64
	TA	7.99	86.27
	SOL	22.52	42.79
V2	MG	12.97	59.96
	TA	13.82	72.15
	SOL	3.45	41.10
V3	MG	9.39	52.64
	TA	11.52	76.36
	SOL	12.21	36.57
V4	MG	8.95	61.11
	TA	6.15	88.09
	SOL	18.01	81.51
V5	MG	3.57	84.47
	TA	11.87	92.36
	SOL	10.21	59.98
V6	MG	21.64	43.79
	TA	4.57	89.14
	SOL	3.87	64.42
V7	MG	6.41	25.90
	TA	3.21	74.75
	SOL	1.28	68.09
V8	MG	0.55	39.96
	TA	2.78	95.80
	SOL	17.99	24.30
V9	MG	12.71	77.57
	TA	6.39	82.16
	SOL	6.10	84.42
V10	MG	23.55	38.30
	TA	3.75	93.56
Overall Average		9.30 ± 6.40	65.39 ± 21.32

matrix H can have different levels of correlations with the joint torque, that is, H_1 is highly correlated with the plantar-flexor joint torque and H_2 is highly correlated to the dorsi-flexor joint torque. This suggests that H_1 and H_2 may be the dual types of neural commands inherently delivered by the CNS for potential control of ankle joint torque. The analysis of synergy intensities could be further applied to FES to control cocontraction levels along with the joint torque control for ankle joint movement. We may stimulate the plantar- and dorsi-flexor muscles simultaneously under different pulsewidth levels to control the ankle joint torque and the joint stiffness by adjusting the coactivations of the plantar-flexor and dorsi-flexor actuators.

From Fig. 4, if we intuitively observe the individual muscle activation (EMG) aspect again, both the SOL and MG activations are highly correlated to the plantar torque. Without synergy extraction, there is no systematic way to quantitatively evaluate the contribution ratios among the agonist muscles or the natural cocontraction usage with the antagonist muscles. For this reason, muscle synergy extraction is essential to the proposed framework, as individual observation of the muscle activation may be insufficient for finding the natural correlation between muscle activations. From the muscle synergies obtained from the ten subjects, as shown in Fig. 3, it appears quantitatively that SOL muscle is the main agonist muscle during joint plantar flexion, whereas MG muscle is not. Considering actual activation state of each muscle, fatigue tracking and evaluation were performed for FES-induced muscle modeling and control [19]. As a future direction toward advanced neuroprosthetics [27], the information from able-bodied subjects on the muscle synergy that is the inherent physiological motor pattern naturally assigned by the CNS should be applied to design multimuscle cocontraction-aware FES systems to conduct different joint flexors and stabilize joint movement.

With regard to estimation performance, the synergy-space method is as accurate for inversely estimating muscle activations as the multiple regression method [22], [23], as shown in Tables I and II. In the isotonic condition, we compare the performances with and without information on joint position and velocity in order to see its impact. Despite no significant difference between the RMSEs (p-value > 0.05 by ANOVA), we suggest adding joint position and velocity information into the inverse estimation, given the classic muscle-length and muscle-velocity relationship [25]. The estimation without joint angle and velocity information is still somewhat accurate, but we suspect that this is due to small range of motion of the ankle joint. Therefore, we assume that joint position and velocity should normally be taken into account for most joints. The effect might be larger than the case we presented. Although it would be quite unrealistic to noninvasively measure the real muscle length and its velocity, joint position and velocity, which reflect the muscle length and velocity relationship, can easily be obtained. Indeed, this assumption is widely used in FES modeling [28]. As we hope to achieve neuroprosthetic muscle control in the future by using averaged muscle synergy information, it would be interesting to see whether we can apply the average of the synergy information obtained from others to a specific subject. Table III shows the performance results of inverse estimation if we replace subject's normalized synergy ratio with the average one from nine other subjects. From the table, we see that the performance is somewhat degraded but acceptable. This implies that there may be some similarities in muscle synergy patterns among the able-bodied subjects. It suggests that, we may be able to apply others' average synergy patterns to an unknown subject for neuroprosthetic control purposes by taking advantage of natural and optimal muscle controls from the intact CNS system. The result in Table III can support the idea of using the generalized muscle synergy pattern to reproduce natural muscle combinations in the patient by means of neuroprosthetics. A recent study [11], in which a patient with complete spinal cord injury was able to regain some motor functions by application of electrical stimulation to the spinal cord, has started to attract considerable attention. In their study, they applied machine learning to help optimally design stimulation patterns. Computer-based intelligent technologies have the potential to play a more active role in rehabilitation applications and in promoting the quality of life for motor-disabled individuals.

The individual relationship between a single muscle and joint torque is worth investigating from a physiological perspective. Fig. 6 shows the individual relationships between predicted

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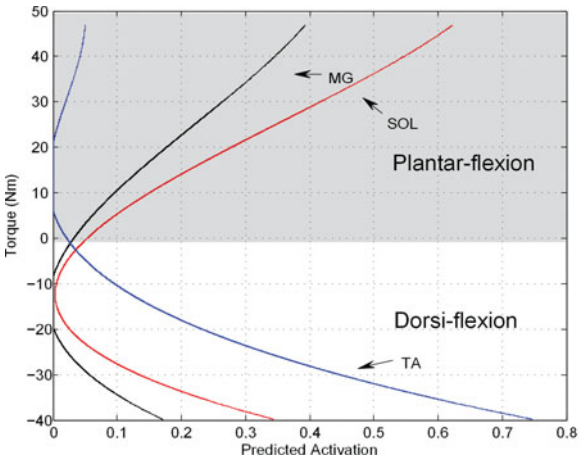


Fig. 6. Relationship between predicted activations and joint torque for each individual muscle: SOL, MG, and TA: the shadow area corresponds to the plantar-flexion situation and the light area corresponds to the dorsi-flexion situation.

activations and torque via the proposed approach (take Subject V1 as an illustrative example). The relationship between muscle activation and joint torque is nonlinear, as the joint torque can be mapped nonlinearly to the individual muscles (SOL, MG, and TA) with polynomial-based models, each with their associated coefficients. This implies a nonlinear relationship between muscle activation and torque, which is consistent with the perspectives pointed out in [25]. As shown in Fig. 6, different cocontraction patterns can occur in plantar flexion and dorsi flexion of the ankle joint [20]. Within the joint torque range $[0, 50\ \mathbf{N}\cdot\mathbf{m}]$ SOL and MG act as agonist muscles for plantar flexion, whereas TA acts as the agonist muscle for dorsi flexion within joint torque range $[-40{,}0\ \mathbf{N}\cdot\mathbf{m}]$. Another interesting finding is that antagonist muscle activation is increased, when the total agonist muscle activation goes into high level, probably to stabilize the joint as the joint becomes unstable at high-contraction state.

V. Conclusion

In this paper, a synergy-based computational framework for the inverse estimation of multiple-muscle activations is proposed and evaluated with experimental analysis. In this first trial, experimental validation is conducted with the ankle joint in isometric and isotonic conditions, and the estimation of three muscle activations (SOL, MG, and TA). The synergies extracted by NMF helped to reduce the dimension to express the muscle activation patterns, the extracted synergies in ten subjects indicated both subject specificity and similarity regarding the multiple muscle activation patterns. The model was cross validated on the prediction of multiple-muscle activations with ten able-bodied subjects in order to evaluate the performance of this method. The approach was found to be promising for inverse estimation of the muscle activities expected from joint torque, both in isometric and isotonic contraction. Regardless of dimension reduction, the performance was not degraded compared with the result with the multiple regression. The statistical analysis showed the significant similarity between methods. This suggests the usefulness of muscle synergy for inverse solutions as it can keep same prediction performance with natural muscle coordination information and in a reduced dimension space operation. In addition, the other one's averaged synergy ratio was applied for muscle activation estimation with leave-one-out cross-validation manner, which resulted in 9.3% estimation error over all the subjects. This result supports the common muscle synergy-based neuroprosthetics control concept.

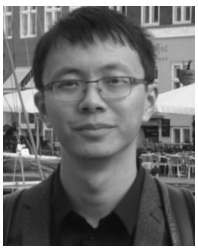
In future work, we intend to investigate the inverse problem in the multiple joints situation, with the ultimate aim of achieving whole-body muscle activity estimation. The extracted information in this situation would be undoubtedly useful for the control of neuroprosthetic systems. The typical use of multiple muscles in able-bodied subjects would be reproduced for the artificial control of the muscles with multichannel FES assisted by the extracted synergy information. The reproduced cocontractions would stabilize the joint, similar to the case of the natural motion employed by CNS.

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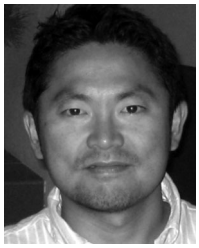
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